Group Forecast and Furious
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PARTITION STRATEGY

The strategy of data partitioning between training, validation, and test datasets was to follow the 60/20/20 rule advised in academic articles (Muraina), as well as other courses of the Applied Analytics program. This rule states that generally, 60% of the entire dataset should be used for training, 20% for model validation, and 20% as a test dataset. This allows the majority of the dataset to be devoted to model training, a sizable portion to evaluate possible models, and another sizable portion to act as a never-before-seen dataset.

The training section of the sentiment dataset encompassed January 2018 through October 2022, resulting in 15,474 rows. The validation set covered November 2022 through October 2023 (7,598 rows) and the test dataset covered November 2023 through October 2024 (6,563 rows). Not every date contained a news article about the AAPL stock, and some dates also contained duplicates, hence the differing number of observations over time periods of the same length.

The AAPL stock price dataset was also partitioned using the same date cutoffs. The resulting datasets had 1,217 rows (training), 251 rows (validation) and 252 rows (test). The sentiment datasets were then aggregated so that each row contained one day and the average sentiment polarity in news articles for that day. These dataset was joined on the stock price datasets to achieve cohesive training, validation, and test datasets.

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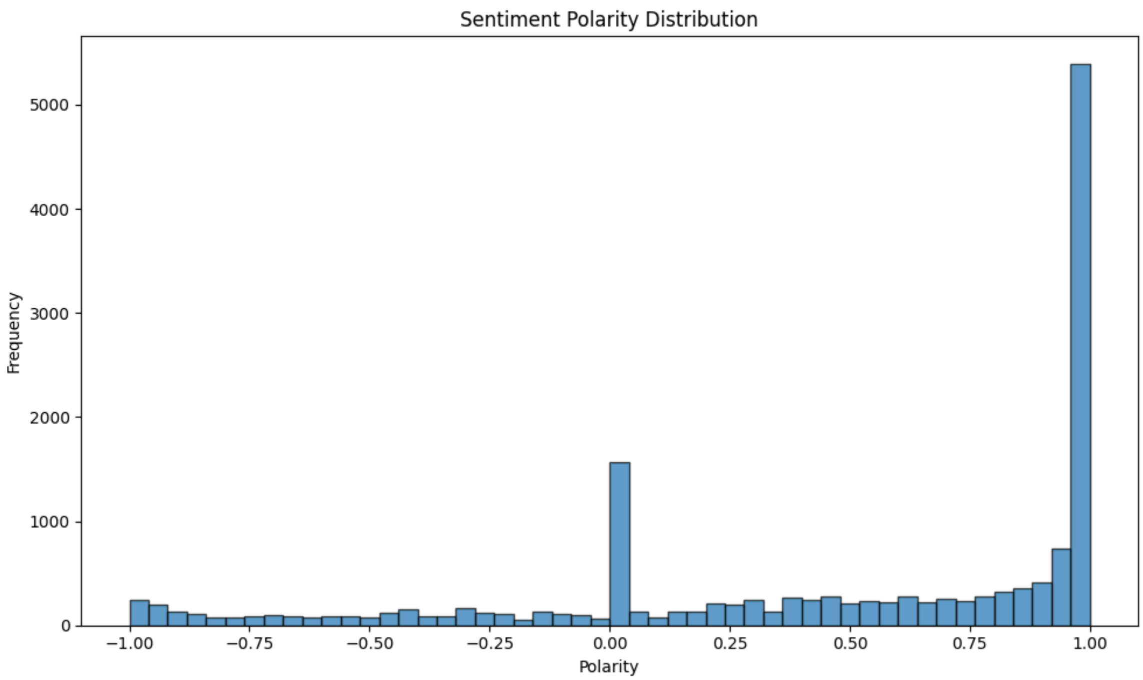
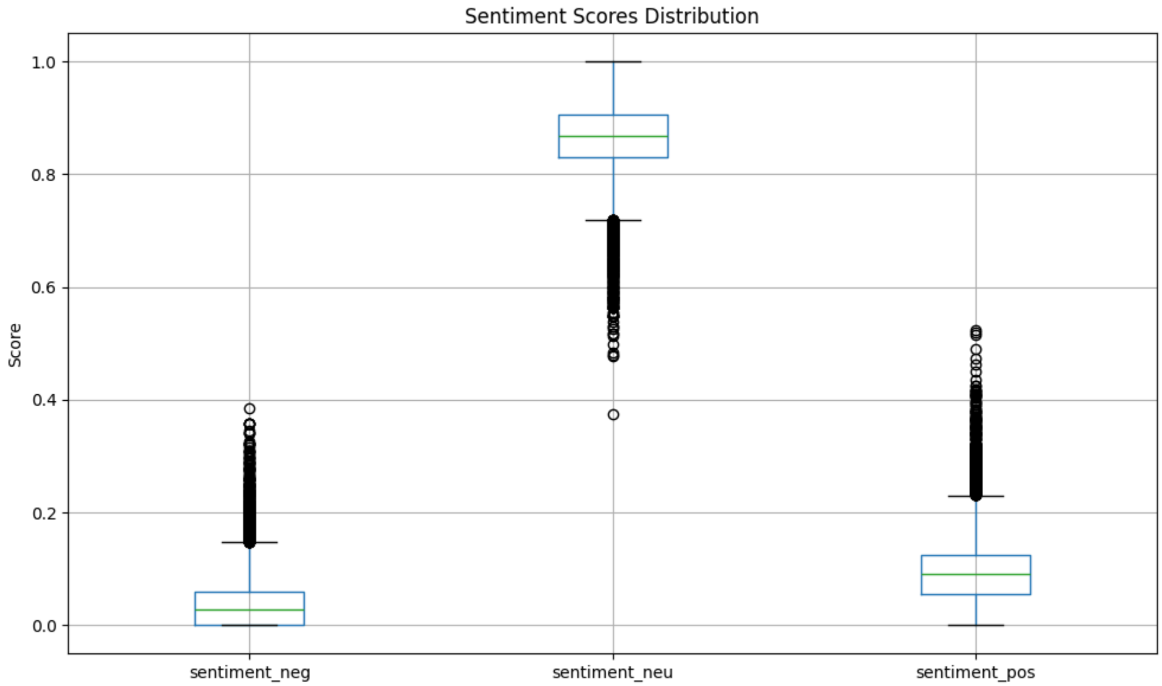
EXPLORATORY DATA ANALYSIS

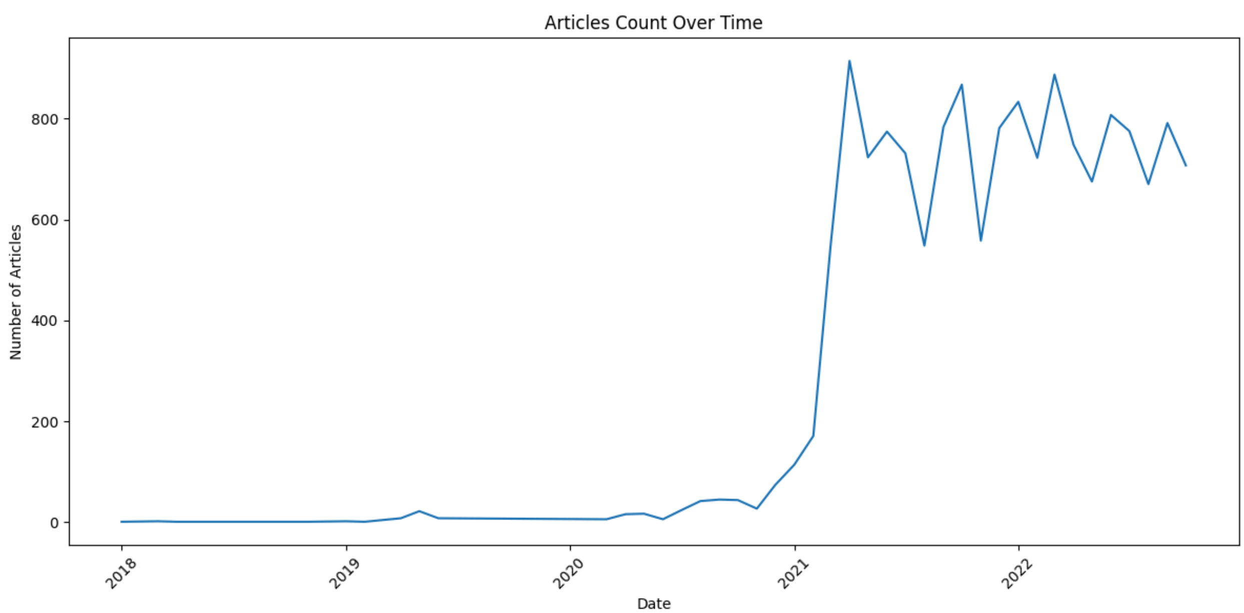
The first section of exploratory data analysis examined the distribution of sentiment polarity within the news sentiment dataset. There was a cluster of observations exactly at 0, marking a perfectly neutral sentiment, while there was an even bigger cluster of observations exactly at 1, marking a perfectly positive sentiment. The rest of the polarities were distributed more evenly, although positive sentiments were slightly more popular, which makes sense given the positive trend of the stock.



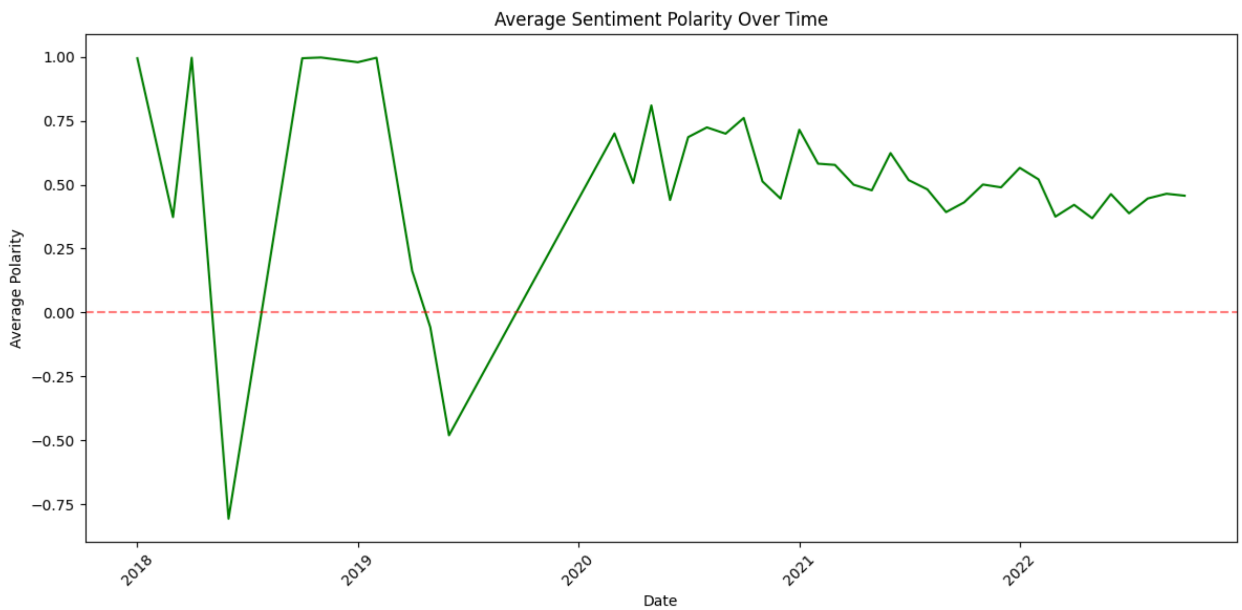
Next, boxplots of the percent negative, percent neutral, and percent positive sentiment scores were generated. The median of percent neutral was close to .9, meaning that approximately 90% of words within the news content was not sentimentally charged. The percent positive median was slightly higher than the percent negative median, indicating there is slightly more positive language present than negative. The inter-quartile distribution for all sentiments was roughly equal.



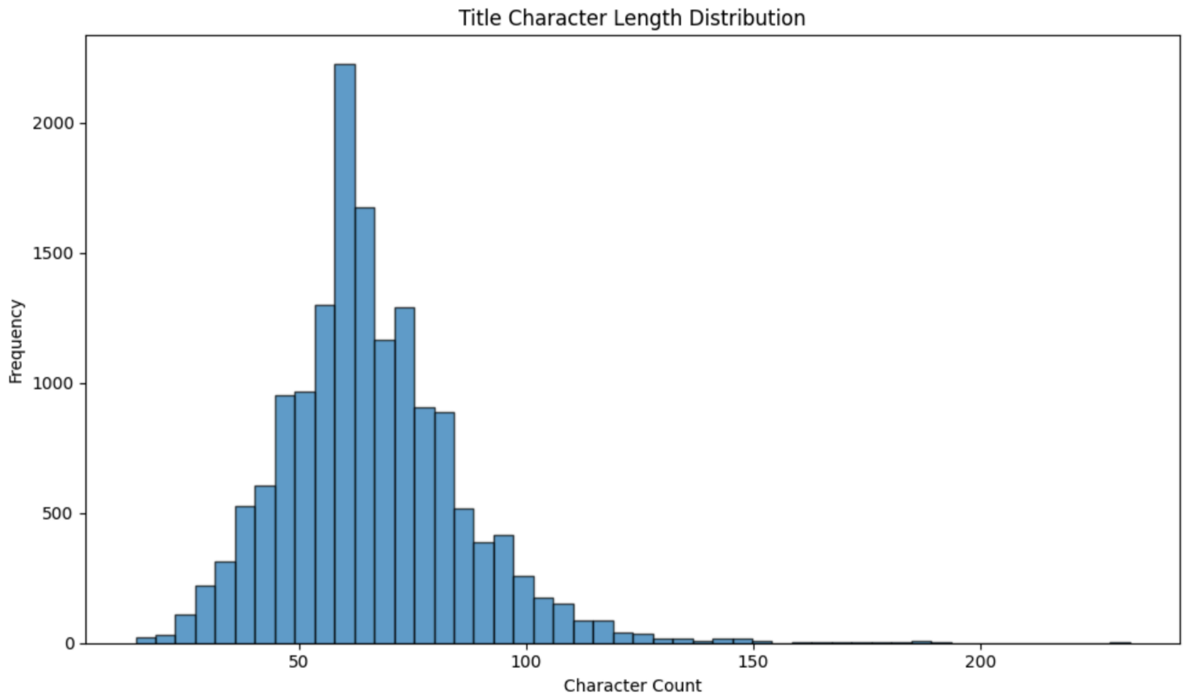
Then, the number of articles in the sentiment dataset was plotted in a simple timeseries. The number of articles increased significantly in 2021 and reached a peak approximately three months into the year. Article count then fluctuated, before stabilizing towards the end of 2022 at a slightly lower article number.

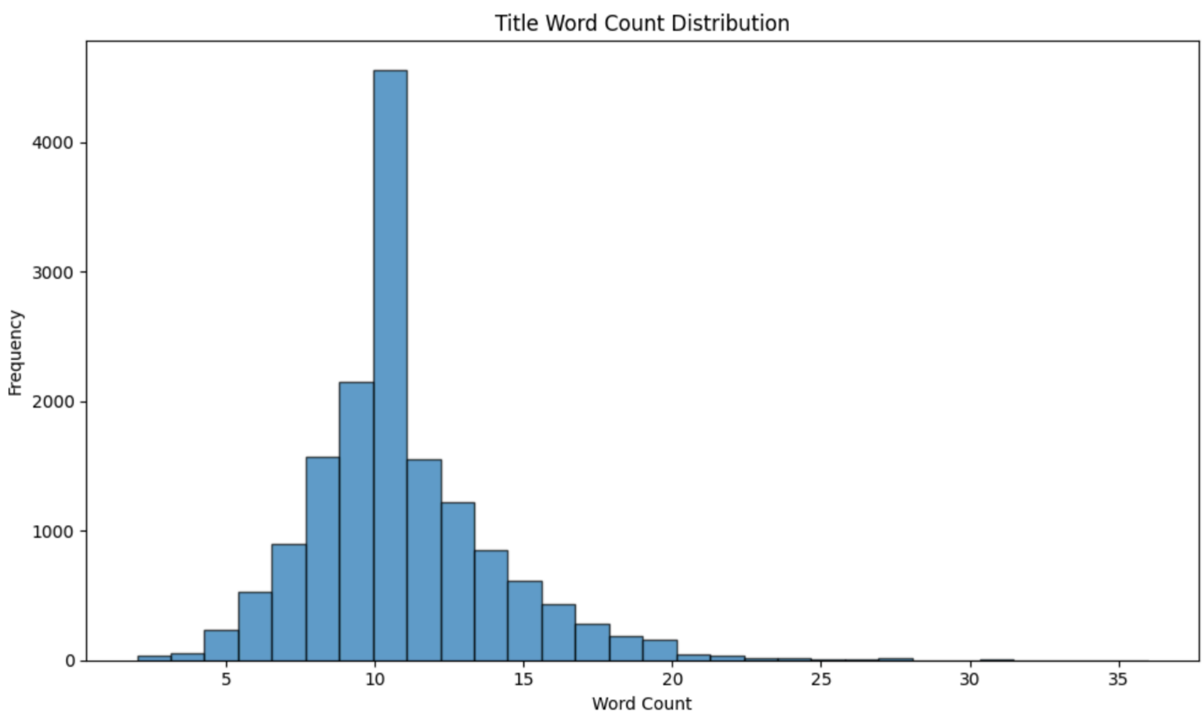


Similar to article count, average sentiment polarity was plotted over time. There was sparse data available from 2018 through 2020, which led to some sharp fluctuations. However, sentiment polarity stabilized around .70 midway through 2020, and slightly declined to around .50 by October 2022. Average positive sentiment polarity was consistently maintained throughout this time period. To identify large periods with missing data, a simple scatterplot was created with date and missing value (True/False). The majority of missing data occurs in 2018 and 2019, but there are stretches edging into 2021 where there are gaps in article coverage.



Next, title length was both plotted in number of characters and words. The title length followed an approximate normal distribution, and the most common frequency was approximately at 60 characters. There was also a slight right skew, where some articles had very long titles in excess of 150 characters Word length followed a similarly normal distribution, where the highest frequency occurred at eleven words. There was a similar right skew where a few outlying titles had upwards of twenty-five words. Within those titles, the most common words were “apple”, “to”, and “stocks”.

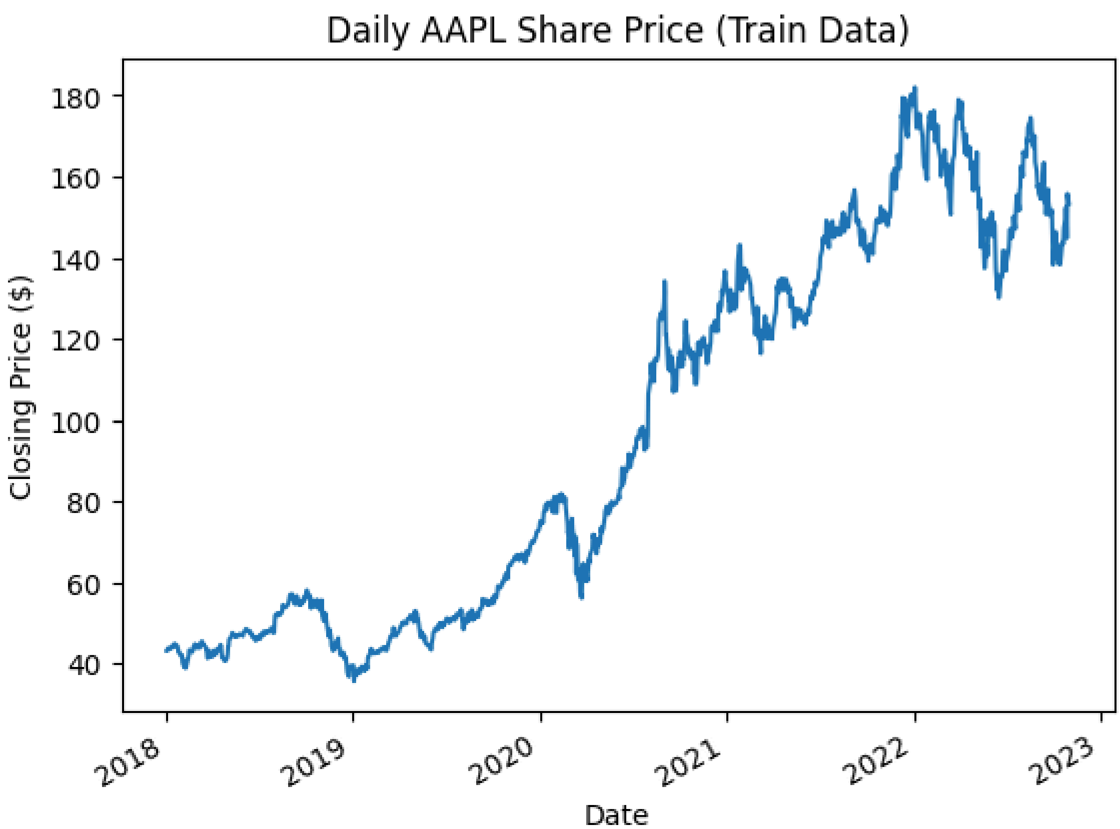




Similar to title length, article content length was also plotted by both character and word length. The character length had an interesting bimodal distribution as many articles had under one thousand characters. However, there was another peak around three thousand characters, although these articles were not as frequent. There was also a severe right skew as a few outlying articles had over twenty thousand characters. World length followed a very similar distribution with the most common word counts under one hundred and around five hundred words. It should be noted that even among these articles under one hundred words, none of them were completely blank or null.

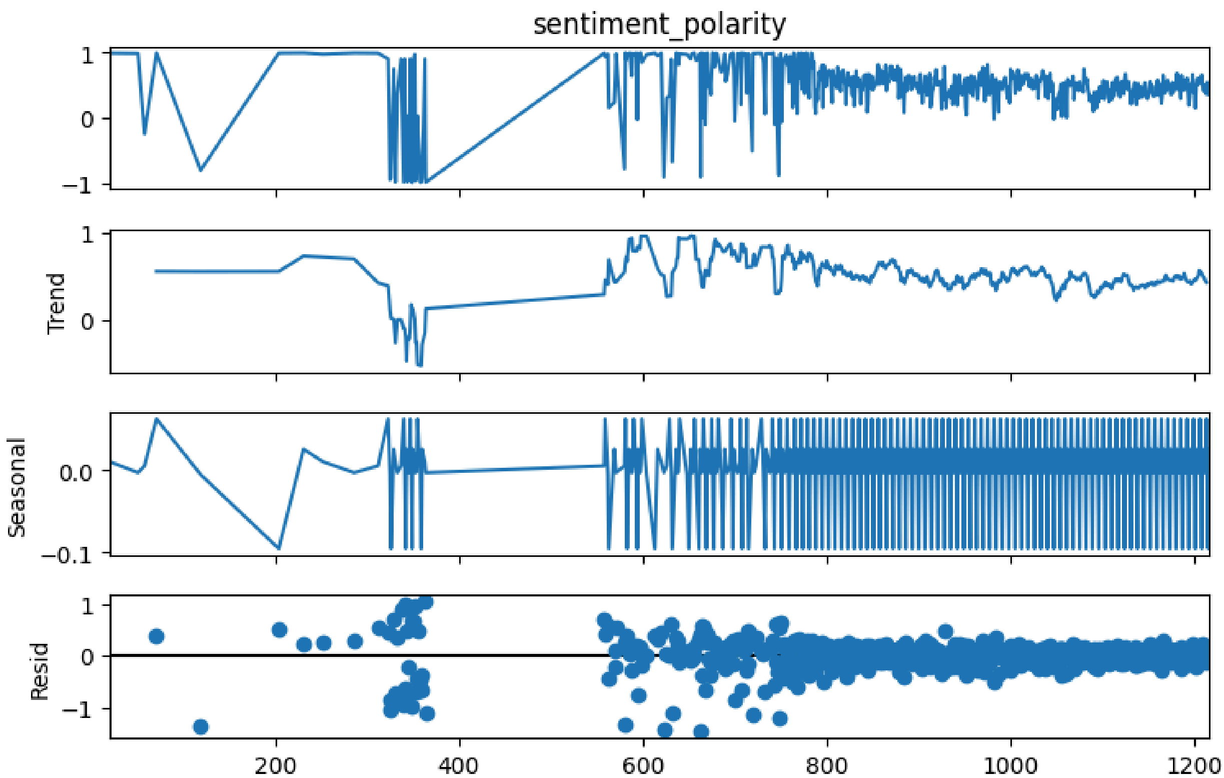
Next, sentiment polarity was plotted against title length in a scatterplot. There was no obvious relationship between the two variables, the points look randomly scattered. Sentiment polarity was also plotted against content word count. The relationship looks to be mostly random, but articles with negative scores, when they exist, tend to be between approximately one hundred fifty and one thousand words of content.

Moving now to the stock price dataset combined with the daily sentiment scores, AAPL stock price was plotted with a simple timeseries. Prices generally increased over time, reaching a peak in January 2022, before decreasing slightly through the end of the year. There were peaks and troughs throughout the entire timeseries, but the most notable increase occurred starting in March 2020. This lines up extremely well with the COVID-19 pandemic, and the stock faltered only slightly when the labor market started to condense at the end of 2022.



Next, a Pearson correlation coefficient was calculated between AAPL Share Price and sentiment polarity, and the resulting value was 0.067. This suggests some correlation between the two variables, but at a very mild level.

Finally, a seasonal decomposition of the timeseries data was calculated. There is a substantial trend component, which shows a slight decrease from a peak around late 2020. There is minimal seasonality, and the residuals are slightly more varied in earlier years, which may be partially due to the lack of consistent available data during this time period.



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ANSWERING THE PROBLEM STATEMENT

Exploratory data analysis absolutely helped to uncover early findings on the problem statement. One intriguing finding was the comparison of average sentiment polarity of time to the AAPL stock price over time. Sentiment was highest in late 2020 and early 2021 when the AAPL stock was surging, and it dropped off (while still positive) when the stock dipped slightly towards the end of 2022. There is an obvious correlation between the two, and this was confirmed with the Pearson correlation coefficient done at the end of the analysis. This value was 0.067, which indicates only a very mild correlation, but it still hints that one exists. This finding bolsters our initial idea to develop three models, of which the later two both use sentiment polarity as a predictor.

Another idea this exploratory data analysis helped develop is a potential mutual dependence relationship between AAPL stock price and sentiment polarity. In other words, news articles may actually be written about the AAPL stock price because it is doing so well (or poorly), not the other way around. It may be difficult to tease apart which variable is affecting the other.

Another idea the timeseries charts within EDA helped to develop is the addition of lags to both the sentiment variable and the stock price direction variable. Stock price direction will not only depend on the sentiment of news articles *today*, but it may also depend on the sentiment of news articles *yesterday* or earlier in the week. The stock price direction may also depend on the direction of previous days, which will be particularly helpful for selecting variables in the final model.

Another insight gained as a result of the exploratory data analysis is the overwhelming majority of sentiments within the articles were positive. This is an important consideration for model development since there is a potential for the model to undervalue the minority negative sentiments during training. Weights for the negative sentiment class may need to be increased, and accuracy metrics such as precision, recall, and F1 score should be used to evaluate the models.

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DATA PROBLEMS

This exploratory data analysis uncovered a few data problems. The first is a massive amount of missing sentiment data between 2018 and 2019. These missing values will clearly need to be handled, either starting the dataset in 2020, imputing the missing values based on previous values, or setting the values to zero to represent a completely neutral sentiment.

The second data problem is an alarmingly high number of articles with less than one hundred words of content. One hundred words is not a substantial amount of content (approximately equal to this single paragraph), so it remains to be seen whether this is enough words to clearly convey a positive or negative sentiment. However, these articles all had at least one word of content, they were not blank. Further analysis should decide whether these articles have enough information to be included in the final combined dataset with AAPL stock prices.

The third data problem is the alarmingly high number of articles with a sentiment polarity nearly at one. The number of articles with a score above .95 dwarfed all other bins with over five thousand articles falling into this category. It may be true that articles are so overwhelmingly positive, especially if they were created during the boom of 2020-2021, but content from a few of these articles should be queried to ensure the validity of these observations.

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REFERENCES

Ismail Olaniyi Muraina. "IDEAL DATASET SPLITTING RATIOS in MACHINE LEARNING ALGORITHMS: GENERAL CONCERNS for DATA SCIENTISTS AND..." *ResearchGate*, unknown, 3 Feb. 2022, www.researchgate.net/publication/358284895_IDEAL_DATASET_SPLITTING_RATIOS_IN_MACHINE_LEARNING_ALGORITHMS_GENERAL_CONCERNS_FOR_DATA_SCIENTISTS_AND_DATA_ANALYSTS. Accessed 6 Feb. 2026.